Algebraic Verification of Two Global Hypergeometric Inequalities

Curvature Certificates and Bernoulli Laplace Envelopes

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Abstract

We give an independent algebraic reconstruction and a complete sign analysis of the two global inequalities used in the exact finite-exchangeability Hoeffding constant. The first is a uniform bound on variance under every exponential tilt of a hypergeometric law. The second is a scaled Bernoulli log-Laplace comparison valid for both population parities in the canonical triangular parameter regime. Exact polynomial identities are checked through a separate symbolic computation and direct finite-sum calculations. A theorem-level concordance fixes the attribution of structural reductions and three-mass proxy formulas.

1 The all-tilt curvature bound

For $H \sim \mathrm{Hyp}(N,k,m)$, write $\mathbb{E}_t H = u(t)$ and $\mathrm{Var}_t(H) = v(t)$ for the exponential tilt proportional to e^{tH}.

Theorem 1.1 (Universal curvature inequality)**.** *For $N \geq 2$, $1 \leq k, m < N$ and $t \in \mathbb{R}$,*

$$v(t) \leq \frac{\min\{k(N-k), m(N-m)\}}{4(N-1)}. \tag{1}$$

Proof. The complement and transposition symmetries reduce to

$$k = j, \qquad m = j+h, \qquad N = 2j+h+r, \qquad j \geq 1, \quad h, r \geq 0.$$

The generating polynomial is

$$Q_j(z) = \sum_{i=0}^{j} \binom{j}{i} (j+h)_{\underline{i}} \frac{(r+j)!}{(r+i)!} z^i.$$

Its Jacobi representation $Q_j(z) = j!(1-z)^j P_j^{(r,h)}((1+z)/(1-z))$ has exclusively negative simple zeros. Under each positive tilt the associated random variable is a sum of j independent Bernoulli variables. Consequently

$$0 < u < j, \qquad 0 < v \leq \frac{u(j-u)}{j}. \tag{2}$$

The polynomial satisfies

$$z(1-z)Q_j'' + [r+1+(j+m-1)z]Q_j' - jmQ_j = 0.$$

Set $z = e^t$. If $v'(t) = 0$, logarithmic differentiation and elimination of z yield $F(u,v) = 0$, where

$$\begin{aligned} D_0(u,v) &= (j-u)(m-u) - v(j+m-2u-1), \\ F(u,v) &= [D_0 - v(r+2u)](v+u^2) + rD_0 u \\ &\quad + (j+m)v(r+2u)u - jmv(r+2u). \end{aligned}$$

As a polynomial in v, F has leading coefficient $-(N-1)$ and positive constant term

$$F(u,0) = u(j-u)(r+u)(m-u) > 0.$$

There is exactly one positive root $v_+(u)$. Set

$$T = N-1, \quad L = N-j-1, \quad D = h-r, \quad S = h+r = L-j+1, \quad B = \frac{j(L+1)}{4T}.$$

At a global maximizer v_*, suppose $v_* > B$. Such a maximizer exists because the tilted variance tends to zero at both ends of the real line. By (2), with $x = 2u_* - j$,

$$A := j^2(j-1) - Tx^2 > 0.$$

An unrestricted polynomial identity gives

$$-16TF\left(\frac{j+x}{2}, B\right) = A(D-x)^2 + (L+1)(2jDx + Ex^2), \qquad E = L(L+1) - 2j. \tag{3}$$

For $j \geq 3$ one has $L \geq j-1$, $E \geq 0$, and $|D| \leq S$. If $xD \geq 0$, the right side is nonnegative. For $xD < 0$, set $q = |x|$ and $d = |D|$. The right side is

$$g(d) = A(d+q)^2 + (L+1)(Eq^2 - 2jqd), \qquad 0 \leq d \leq S.$$

If $A \geq j^2/L$, completion of squares gives

$$g(d) = A\left(d + q - \frac{j(L+1)q}{A}\right)^2 + (L+1)q^2\left[E + 2j - \frac{j^2(L+1)}{A}\right] \geq 0,$$

using $E + 2j = L(L+1)$.

Suppose $0 < A < j^2/L$. Then

$$q > q_0 := j\sqrt{\frac{(j-1)L-1}{LT}} \geq \frac{j}{L}.$$

The final inequality follows from $(j-1)L^2 - 2L - j \geq 0$ for $L \geq j-1$, $j \geq 3$. Hence

$$A(d+q) \leq \frac{j^2}{L}(S+q) \leq j(L+1)q,$$

so g decreases on $[0, S]$. Write $a = j-3 \geq 0$, $L = j-1+S$. The inequality $Eq_0 \geq 2jS$ is equivalent to the polynomial nonnegativity

$$\mathcal{K} := E^2((j-1)L-1) - 4S^2L(L+j) \geq 0. \tag{4}$$

Its exact expansion is

$$\begin{aligned}\mathcal{K} = {} & a^2(a+1)(a+3)^3 + a(a+3)^2(5a^2+16a+10)S \\ & + (10a^4+80a^3+213a^2+202a+35)S^2 \\ & + (10a^3+64a^2+117a+52)S^3 \\ & + (5a^2+22a+19)S^4 + (a+2)S^5.\end{aligned}$$

Every coefficient is nonnegative for $a, S \geq 0$. Therefore $g(d) \geq g(S) \geq 0$, and (3) gives $F(u_*, B) \leq 0$. The positive root $v_+(u_*)$ lies below B, a contradiction.

For $j = 1$, the tilted law is Bernoulli and $v \leq 1/4 = B$. For $j = 2$ the centered generating polynomial is proportional to $1 + 2\rho e^s + e^{2s}$, where

$$\rho = \sqrt{\frac{m(N-m)}{(m-1)(N-m-1)}} \in \left[\frac{N}{N-2}, 2\right].$$

Writing $q = \cosh s \geq 1$ gives

$$v = \frac{\rho q + 1}{(\rho+q)^2} \leq \frac{1}{1+\rho} \leq \frac{N-2}{2(N-1)} = B.$$

All parameter cases follow by symmetry. □

2 The mean–variance Laplace comparison

Theorem 2.1 (Parity-uniform scaled Bernoulli comparison)**.** *Let $N \geq 7$, $3 \leq m \leq k \leq \lfloor N/2 \rfloor$ and $N > 2m$. Set $p = k/N$, $M = m(N-m)/(N-1)$ and $a = (N-2m)/(N-2)$. For every $t \in \mathbb{R}$,*

$$\log \mathbb{E}e^{t(H-mp)} \leq \frac{M}{a^2}\left[\log(1-p+pe^{at}) - pat\right], \tag{5}$$

with strict inequality for $t \neq 0$.

Proof. Put $\mu = mp$, $d = 1-2p$ and $u = \mathbb{E}_t H$. Since the tilted mean increases strictly from 0 to m, its variance defines $V(u)$. Real-rootedness gives $V(u) \leq u(m-u)/m$.

Write

$$A(u) = u(u+N-k-m), \qquad C(u) = (k-u)(m-u),$$
$$F(u,v) = -(N-1)v^2 + [A + C - A'C + C'A]v + AC.$$

The hypergeometric adjacent-mass recurrence yields the exact differential equation

$$F(u,V) + N(u-\mu)VV' = 0. \tag{6}$$

Introduce the quadratic

$$P(u) = Mp(1-p) + ad(u-\mu) - \frac{a^2}{M}(u-\mu)^2$$

and $z_0 = 2(u-\mu)/M$, together with

$$G = \frac{N^2(m-1)(N-m-1)}{4(N-2)^2(N-1)} > 0,$$
$$W(d,z_0) = N - 4 + 3Nd^2 - 4Nadz_0 + [(N+1)a^2 - M]z_0^2.$$

Polynomial expansion in u, k, m, N gives

$$F(u,P) + N(u-\mu)PP' = -G(u-\mu)^2 W(d,z_0). \tag{7}$$

Let $\delta = (N-2m)/N$, $b = (N-m)/(N-2)$ and

$$c = \frac{(N-m)[3N^2 - 4N - 4m(N-1)]}{(N-1)(N-2)^2}, \qquad E(d,z_0) = 1 - d^2 + 4bdz_0 - cz_0^2.$$

The exact admissibility identity is

$$\frac{u(m-u)}{m} - P(u) = \frac{m(m-1)}{4(N-1)}E(d,z_0). \tag{8}$$

Thus at a contact $V = P$, one has $E(d,z_0) \geq 0$. Also $0 \leq d \leq \delta$.

Define

$$\lambda = \frac{m^2 - mN + 3m + N^2 - 4N}{2(N-m)}, \qquad \alpha = \frac{N-4-\lambda}{\delta^2},$$
$$A_0 = 3N + \lambda + \alpha, \quad B_0 = -2Na - 2\lambda b, \quad C_0 = (N+1)a^2 - M + \lambda c.$$

A second exact identity gives the quadratic certificate

$$W = \lambda E + \alpha(\delta^2 - d^2) + A_0 d^2 + 2B_0 dz_0 + C_0 z_0^2. \tag{9}$$

Its determinant factors as

$$A_0C_0 - B_0^2 = \frac{N^2 U_0 V_0}{(N-2m)^2(N-2)^2(N-1)}, \tag{10}$$

where $U_0 = N^2-2N-m^2-mN+3m$ and $V_0 = m^3-6m^2-2mN^2+14mN-3m+N^3-7N^2+6N$. Write $j = m-3 \geq 0$, $r = N-2m \geq 1$. Direct expansions are

$$\begin{aligned} 2(N-m)\lambda &= 3j^2+3jr+13j+r^2+5r+12, \\ 2(N-m)(N-4-\lambda) &= j^2+3jr+3j+r^2+5r, \\ U_0 &= j^2+3jr+5j+r^2+7r+6, \\ V_0 &= j^3+4j^2r+3j^2+4jr^2+10jr+r^3+5r^2. \end{aligned}$$

Each expression is positive on the indicated integer domain. Therefore $\lambda, \alpha, A_0 > 0$ and the quadratic form in (9) is positive definite. In particular, $W(d, z_0) > 0$ at every contact $V = P$.

At $u = 0, m$ the quadratic barrier P is positive, while V vanishes. At $u = \mu$, the two curves meet. If $f = V - P$ vanishes at any other interior point, (6) and (7) yield

$$f'(u) = \frac{G(u-\mu)W(d, z_0)}{NP(u)}.$$

Every noncentral zero to the left of μ crosses downward, and every such zero to the right crosses upward. The endpoint signs exclude a positive component of f on either side. The same derivative formula excludes further interior equality. Thus $V(u) < P(u)$ for $u \neq \mu$.

Finally, put $y(t) = u(t) - \mu$. The quadratic differential inequality is

$$y' \leq Mp(1-p) + ady - \frac{a^2}{M}y^2.$$

Its exact equality solution with initial value zero is

$$Y(t) = \frac{M}{a}\left(\frac{pe^{at}}{1-p+pe^{at}} - p\right).$$

Variation of constants gives $y(t) < Y(t)$ for $t > 0$ and $y(t) > Y(t)$ for $t < 0$. Integration with the corresponding orientation yields (5) and the asserted strictness. □

3 Theorem provenance and scope

The sharp weighted finite-exchangeability constant is obtained by inserting Theorems 1.1 and 2.1, together with the $m = 1, 2$ computations, into the hypergeometric variational identity of Lee and Kim [1]. Their Theorem 1 bounds the inflation by $\Gamma_N = 1 + 3/(2N) + O(N^{-2})$; Proposition 1 gives the variance lower bound, Remark 1 states the exact hypergeometric supremum, and Proposition 2 gives the two-level reduction. The present sharp evaluation is

$$C_N^\star = \begin{cases} N/(N-1), & N \text{ even}, \\ [(N-1)\operatorname{artanh}(1/N)]^{-1}, & N \text{ odd}. \end{cases}$$

The three-mass optimal-proxy identity used for $m = 2$ is Theorem 3.2 of Atouani, Marchal and Arbel [2]. The hypergeometric two-draw maximization is an additional optimization over the success count. Classical Jacobi zero localization [6] supplies the real-root factorization. General optimal-proxy characterizations occur in Arbel, Marchal and Nguyen [3]. Barber's exchangeable weighted inequality [4] and the de Finetti extremal-mean concentration of Gottschling and Caprio [5] address related bounds with different constants or hypotheses.

The independent verifier `checks/verify_independent_certificates.py` reconstructs the stationary identity, positivity polynomial, residual equation, quadratic certificate, determinant and sign expansions from their defining expressions. It checks the generating-polynomial differential equation in 96 exact cases and evaluates log-Laplace transforms directly for $7 \leq N \leq 18$. The mathematical implications in the proofs above are uniform in all admissible parameters.

References

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